

**LABORATORY EXPERIMENT LIST**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**GCU, 2024**

**SUBJECT: PROGRAMMING FOR PROBLEM SOLVING LAB**

| Experiment | Experiment Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>I</b>   | <b>BASIC OF C PROGRAMMING:-</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|            | a) Write a program to display you name, rollno,branch and semester.<br>b)Write a program to find:<br>i.     Addition of two numbers.<br>ii.    Subtraction of two numbers.<br>iii.   Multiplication of two numbers.<br>iv.    Division of two numbers.<br>c)Write a program to find area of :<br>i)Rectangle. (ii) Circle.(iii)Triangle.<br>d) Write a program to find simple interest, compoundinterest and amount.                                                                                                                                      |
| <b>II</b>  | <b>CONDITIONAL CONTROL STATEMENTS:-</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|            | a) Write a program to check whether given number is odd or even.<br>b) Write a program to find the greatest among two numbers.<br>c) Write a program to find the greatest of three numbers.<br>d) Write a program to check whether a person is eligible for vote or not.<br>e) Write a program to check whether a given year is leap year or not.<br>d) Write a program to find the area of a triangle, rectangle and circle using switch case.<br>e) Write a program to perform addition, subtraction & multiplication of two numbers using switch case. |
| <b>III</b> | <b>LOOP CONTROL STATEMENT:-</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|            | a) Write a program to calculate the sum of all the numbers from 1 to 50 using for loop.<br>b) Write a program to display your name upto 10 times using while loop.<br>c) Write a program to print even numbers from 1 to 50 using do-while loop.<br>d) Write a program to reverse a given no.<br>e) Write a program to find Fibonacci series.<br>f) Write a program to find factorial of a no.<br>g) Write a program to print all the numbers divisible by 2 and 3 between 1 and 50.                                                                      |
| <b>IV</b>  | <b>ARRAY:-</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|            | a) Write a program to read and display an array of elements.<br>b) Write a program to find position of an element in an array.<br>c) Write a program to addition of two matrixes.                                                                                                                                                                                                                                                                                                                                                                         |

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| <b>V</b>   | <b>FUNCTIONS:-</b>                                                                                                                                                                                                                                                                                                              |
|            | a) Write a program for swapping of two numbers using functions.<br>b) Write a program to find the factorial of a number using function.<br>c) Write a program in C to check whether a number is a prime number or not using the function.<br>d) Write a program in C to check Armstrong and Perfect numbers using the function. |
| <b>VI</b>  | <b>POINTER:-</b>                                                                                                                                                                                                                                                                                                                |
|            | a) Write a program in C to show the basic declaration of a pointer.<br>b) Write a program in C to demonstrate the use of the & (address of) and *(value at address) operators.<br>c) Write a program in C to add numbers using call by reference.<br>d) Write a program in C to read and display array elements using pointer.  |
| <b>VII</b> | <b>STRING:-</b>                                                                                                                                                                                                                                                                                                                 |
|            | a) Write a program in C to input a string and print it.<br>b) Write a program in C to find the length of a string without using library functions.<br>c) Write a program in C to count the total number of words in a string.<br>d) Write a program in C to compare two strings without using string library functions.         |